

西南财经大学本科期末考试试题册（B）

（2021—2022学年第2学期）

以下各项由命题教师填写：

课程名称：面向对象程序设计（JAVA EE）（英）命题教师：陈中普

适用对象（年级专业）：2021级信息管理与信息系统

使用试题的任课教师姓名：陈中普

试题说明：

- 1、考试类型：闭卷[] 开卷[☒]
- 2、本套试题共 三 道大题，共 4 页，完卷时间 120 分钟。
- 3、考试用品中除纸、笔、尺子外，可另带的用具有：
计算器[] 字典[] 等
(请在下划线上填上具体数字或内容，所选[]内打钩)

考试时间（由制卷方填写）：

以下各项由学生填写：

任课教师：

年级专业：

学生姓名：

学 号：

- 考生注意事项：
1. 出示学生证（或身份证）和准考证于桌面左上角，以备查验。
 2. 拿到试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。
 3. 答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名填写完整。
 4. 所有答案均需填写在答题纸上，答在试题册上无效。
 5. 考生不得携带任何通讯工具进入考场。
 6. 考试结束后，将试题册、答题纸和草稿纸等下发材料上交监考教师，不得带离考场。
 7. 严格遵守考场纪律。

Part One: Multi-choices (each carries 2 marks; 20 marks in total)

(Note: each question has only one correct choice.)

1. Which of the following statement is correct in terms of Java EE?
 - A. It is difficult to reuse Java SE code in Java EE project.
 - B. Java EE application runs within in a container.
 - C. The latest Long-Term-Support version of Java SE is 17, so is Java EE.
 - D. Java is the only choice to develop a website.
2. Which of the following statement is correct in terms of HTML?
 - A. `<h2>` renders a bigger text than `<h1>` does.
 - B. Only `<body>` part is visible on the web page.
 - C. `<p><span>hello</p></span>` is nested properly.
 - D. We can use pure HTML to implement a calculator (e.g., a plus b, a div b).
3. Which of the following statement is correct in terms of Servlet?
 - A. The `init()` will be called every time a new request is coming.
 - B. A JSP will be translated into a servlet during runtime.
 - C. It is not legal to map a servlet to URL `"/`.
 - D. We can only output HTML in a servlet.
4. Which of the following statement is correct in terms of HTTP?
 - A. A simple hyperlink always means a GET method.
 - B. We cannot send images through HTTP.
 - C. Most of the time in a payment website, we use GET method.
 - D. The default method of a form is POST method.
5. Which methods are used by a servlet to handle form data from a client?
 - A. `HttpServlet.doHead()`
 - B. `HttpServlet.doPost()`
 - C. `HttpServlet.doForm()`
 - D. `ServletRequest.doGet()`
6. Given a `req` is an `HttpServletRequest`, which snippets create a session if one doesn't exist?
 - A. `req.getSession()`
 - B. `req.createSession()`
 - C. `req.createSession(true)`
 - D. `req.getNewSession()`
7. Which of the following statement is correct in terms of init paramters and attributes?
 - A. One can only set the Servlet init parameters in DD (`web.xml`) file.
 - B. One can set a `Book` object for init parameters.
 - C. Servlet A cannot read Servlet B's init parameters.
 - D. The Attribute is an alias of the init parameter.

8. Which of the following statement is correct in terms of Session?
 - A. Both cookies and the session are stored in the server.
 - B. Suppose Bob visits *JD.com* using his iPhone and Laptop at the same time, there are two sessions.
 - C. We can use *Session.stop()* to kill a session.
 - D. We can only put strings into a session's attributes.

9. Suppose there is an object *user* which has a field "country", and we put it into the attribute in a servlet: *request.setAttribute("user", user)*. Then we forward this request to *test.jsp*.
 - A. We must use "user" as the name of this attribute.
 - B. In *test.jsp*, we can always use *\${user.country}* to get the country of this user.
 - C. We can also use *Redirect* to go to *test.jsp*.
 - D. It would be better to hide this JSP file in *WEB-INF*.

10. Which of the following statement is correct in terms of Java Web?
 - A. A Servlet is often used as a Model.
 - B. Since multiple threads may update a session, it is not thread safety.
 - C. It is a good practice to write Java codes in JSP.
 - D. One servlet can be mapped to several URLs.

Part Two: Short-answer Questions (10 marks + 15 marks + 20 marks)

1. Alice wants to design a web page which enable users to input their favorite beer favors, and then the server can recommend beers given the selection. Suppose the code snippet from servlet is:

```

1. @WebServlet(value="/beers")
2. public class BeerServlet extends HttpServlet {
3.     public void doPost(HttpServletRequest request, HttpServletResponse response) {
4.         String favor = request.getParameter("favor");
5.         ...
6.     }
7. }
```

Please write down the key HTML code of *<form>* element.

```

1. <form ①>
2.     <input ②>
3.     <button type="③">Beer Recommendation</button>
4. </form>
```

2. Alice would like to generate a QR Code in her web app, and this is what she found a Gradle dependency of an awesome library:

```
implementation 'com.google.zxing:core:3.5.0'
```

- a) What can you tell from it?
 - b) If someone claims that “Since we are going to return an image, we must use POST method”, how do you think of it?
3. Suppose a list of users are stored in a request’s attribute, and the following is the code snippet:

```
1. List<User> users = db.getUsersByCountry(country);
2. request.setAttribute("users", users);
3. RequestDispatcher dispatcher = request.getRequestDispatcher("users.jsp");
4. dispatcher.forward(request, response);
```

User here is a regular Java bean, which contains attributes *country* and *name*.

- a) Please write down the key code to display the prices of returned users using JSTL.
- b) Alice thinks it does not make sense to use *request.setAttribute()*, and then call *sendRedirect()*. Try to defend her idea.
- c) By looking up the Java Doc, Alice knows the second parameter of *setAttribute()* method is an *Object*. Please summarize why Java EE’s designers choose the *Object* as the parameter type.
- d) Suppose we find the following CSS code in *users.jsp*. What does *h1.name* mean?

```
1. h1.name {
2.     text-align: center;
3. }
```

Part Three: Design (10 marks + 25 marks)

1. Suppose the home page of the Java EE course is <https://it.swufe.edu.cn/undergraduate/courses/CS63020>, where CS63020 is the course code. One day the teacher decided to develop a new website for this course by adding many new features. What’s more, a fancy and easy-to-remember URL <https://javaee.swufe.org/> is used, which renders the previous one obsolete. However, many people have already added the previous one in their bookmarks, so it is likely for them to visit an old and obsolete website.

Please design a solution to this problem.

2. Suppose your teammate has completed the logic for insert a new user into database, and he offers you a public API in *User* class which has a non-arg constructor:

```
1. /// Return true if added successfully;  
2. /// Otherwise return false (e.g., the email exists).  
3. public boolean addUser(String email, String country, String password)
```

You shall use it to insert a user. Please complete the code in a servlet based on the process flow. Note that the country can be in non-Latin characters.

Start

```
1. ⑥  
2. String email = request.getParameter("email");  
3. String password = request.getParameter("password");  
4. String country = request.getParameter("country");  
5. // Is the insertion successful?  
6. if (⑦) {  
7.     // add to /login  
8.     ②  
9. } else {  
10.     ③  
11. }
```